

Ginseng Protein Reverses Amyloid Beta Peptide and H₂O₂ Cytotoxicity in Neurons, and Ameliorates Cognitive Impairment in AD Rats Induced by a Combination of D-Galactose and AlCl₃

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Ginseng (*Panax ginseng* C.A. Meyer) is one of the most widely used herbal medicines worldwide. The present study evaluated the neuroprotective effects of ginseng protein (GP) and its possible mechanisms in a cellular and animal model of AD. The results demonstrated that GP (10–100 µg/mL) significantly improved the survival rate of neurons and reduced the cells' apoptosis and the mRNA expression of caspase-3 and Bax/Bcl-2. In addition, GP (0.1 g/kg) significantly shortened the escape latency, prolonged the crossing times and the percentage of residence time; reduced the level of Aβ_{1–42} and p-tau, the activity of T-NOS and iNOS, and the content of MDA and NO, improved the activity of SOD, the concentration of cAMP and the protein expression of p-PKA/PKA and -CREB/CREB. The results demonstrated that GP significantly inhibited Alzheimer-like pathophysiological changes induced by Aβ_{25–35} or H₂O₂ in cells or those induced by D-gal/ Al in animals. These neuroprotective effects of GP may be associated with the cAMP/PKA/CREB pathway. Also, in combination with our previous studies, these results indicate that the anti-AD mechanism of GP was likely to activate the CREB pathway through multiple channels. Copyright © 2016 John Wiley & Sons, Ltd.

Keywords: Alzheimer's disease; neuroprotective; ginseng; protein; oxidative stress.

INTRODUCTION

Alzheimer's disease (AD) is an irreversible neurodegenerative disorder clinically characterized by progressive cognitive dysfunction and memory decline. The abnormal deposition of beta amyloid peptide (Aβ) is one of the major neuropathological hallmarks of AD (Koo *et al.*, 2013; Götz *et al.*, 2007; Mattson 2004; Minter *et al.*, 2016; Wimo 2015). Oxidative stress plays a significant role in the pathogenesis of AD (Mawuenyega *et al.*, 2010; Rosini *et al.*, 2013). Aβ, particularly the Aβ fragment 25–35 (Aβ_{25–35}), and H₂O₂ induced neurotoxicity have been widely used for researching the effects of oxidative stress injury on neuronal responses (Trubetskaya *et al.*, 2003; Tran *et al.*, 2001; Alkam *et al.*, 2008; Choi *et al.*, 2003; Rafatian *et al.*, 2012). Over-supply of D-galactose (D-gal) could accelerate the generation of reactive oxygen species (ROS), and subsequently cause oxidative stress

and cellular damage (Peng *et al.*, 2015; Kumar and Gill, 2009; Yang *et al.*, 2013). Aluminium (Al) is a neurotoxic agent that could exacerbate oxidative stress (Peng *et al.*, 2015; Zaky *et al.*, 2013); therefore, combining D-gal with Al has become a useful animal model that has been widely used to study potential treatments for AD (Sun *et al.*, 2009; Honig *et al.*, 2003; van Oijen *et al.*, 2005; Xiao *et al.*, 2011; Luo *et al.*, 2009; Peng *et al.*, 2015; Yang *et al.*, 2014; Yang *et al.*, 2013; Li *et al.*, 2008; Zhang *et al.*, 2014).

Ginseng, the root and rhizome of *Panax ginseng* (C. A. Meyer), has neuroprotective and cognitive-enhancing functions in animals, healthy individuals and AD patients who have been widely reported (Lee *et al.*, 2008; Lee *et al.*, 2009; Heo *et al.*, 2012). Proteins are one of the main components of ginseng (Ru *et al.*, 2015), and those with a molecular weight of 28, 26, 21, 20 and 35.2 kDa have been extracted and purified from *P. ginseng* (Yoon *et al.*, 2002; Kim *et al.*, 2004; Hwang *et al.*, 2001; Moon *et al.*, 2010; Ng and Wang, 2001). We also extracted the ginseng protein (GP) and purified the proteins of molecular weights of 66, 65, 25, 15 and 8 kDa that have been described previously (Li *et al.*, 2016), including the 15 kDa protein that is a superoxide dismutase (SOD), which was as a homodimer reported in our previous article (Li *et al.*, 2010). Our previous

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studies suggest that GP may improve memory ability and reduce the pathological injury of AD rats induced by D-gal with Al, and the neuroprotective effects of GP may be related to the activation of the PI3/Akt pathway (Li *et al.*, 2016), which provided a pharmacological basis for the further study of the anti-AD effect of GP.

In this work, we assessed the effects of GP on survival and apoptosis of neurons *in vitro*, investigated its impact on cognitive functions and pathological injury of AD rats *in vivo*, and subsequently explored its correlation with the cAMP/PKA/CREB signalling pathway to reveal the mechanism underlying the neuroprotective effect of ginseng.

MATERIALS AND METHODS

Materials. Coomassie brilliant blue R-250 from Amresco (USA); BCA protein assay kit from Pierce (USA); coomassie brilliant blue kit, TEMED, MTT, A β _{25–35}, DAPI, Hoechst 33342, propidium iodide (PI) and CyTM 3-conjugated anti-rabbit IgG from Sigma-Aldrich (USA); DMEM high-glucose media, fetal bovine serum (FBS) and Penicillin-Streptomycin from Gibco (USA); rabbit anti-Neurofilament M (NF-M) from StemCell Technologies (Canada); mounting media from Southern Biotech (USA); A β _{1–42} and p-tau Elisa kit from Shanghai Sunred (China); cAMP assay kit from R & D (USA); superoxide dismutase (SOD), malonaldehyde (MDA), total nitric oxide synthase (T-NOS), inducible nitric oxide synthase (iNOS) and nitric oxide (NO) kit from Nanjing Jiancheng Bioengineering Company (China); M-MLV reverse transcriptase from Toyobo (Japan); TRIzol and cDNA Synthesis Kit from Thermo (USA); PCR Master Mix from ABI (USA); ECL detection reagents from Millipore (USA); D-gal and anti-rabbit horseradish peroxidase conjugated (HRP) from Genview (USA); rabbit anti-PKA and PKA (Phospho-Thr 197) from Signalway Antibody (SAB, USA); rabbit anti-CREB and CREB (Phospho-Ser133) from Cell Signaling (CST, USA); rabbit anti tubulin from Abcam (UK). All other chemicals and reagents are of analytical grade.

Determination of molecular weight and purity of GP. *P. ginseng* roots of four-year-old plants were purchased from Changbai, Jilin, China. The plant was authenticated by Professor Dacheng Jiang (College of Pharmacy, Changchun University of Traditional Chinese Medicine, Changchun, China). The voucher specimens were deposited in the Jilin Ginseng Science Research Institute (Jilin, China).

GP was provided by Professor Yu Zhao of Changchun University of Traditional Chinese Medicine using methods that were described previously (Li *et al.*, 2010). Briefly, the *P. ginseng* roots were extracted by 10 mM Tris-HCl (pH 7.4), and the homogenate was stirred for 24 h, and then centrifuged at 10 000 rpm for 10 min. Supernatant was precipitated by ammonium sulfate (80% final concentrations, w/v) and then was centrifuged again. The resultant precipitate was sequentially dialysed in the extraction buffer for 24 h at room temperature (RT). Dialysed liquid was collected and frozen dried to

obtain GP. The protein content was determined by the coomassie brilliant blue protein assay kit using bovine serum albumin (BSA) as a standard with a MR-96A Microplate Reader (Mindray, Shenzhen, China). The molecular weight of protein subunits were separated by sodium dodecyl sulfate-polyacrylamide gel electrophoresis (SDS-PAGE, resolving gel 15% and stacking gel 5%), and the running voltage was 150 V using the 165–8001 mini-protean4 vertical electrophoresis (Biorad, USA). The protein bands were visualized using the coomassie brilliant blue R-250 staining. The molecular weight of native protein was analysed using the 1100 high performance liquid chromatograph (HPLC, Agilent, USA) equipped with UV and refractive index detectors and an Ohpak SB-803 HQ gel column (8.0 × 300 mm, Showa Denko, Japan). The exclusion limit of the gel column was 100 kDa, the mobile phase was 10 mM Tris-HCl (including 10 mM NaCl, pH 7.4) at a flow rate of 1.0 mL/min. The GP concentration was 2 mg/mL (dissolved in distilled water), the sample injection volume was 15 μ L, the column was operated at 25 °C, and the detection wavelength was 280 nm. The molecular weights of the protein marker used to draw the standard curve were as follows (in descending order): 97.4, 66.2, 43.0, 31.0, 20.1 and 14.4 kDa.

Culture and assessment of primary cortical neurons. Pregnant KM mice were provided by the Liao Ning Chang Sheng Biotechnology Co., LTD, China. The experimental animal quality certification permit number was SCXK (Liaoning) 2010-0001.

Primary cortical neurons were isolated and cultured from neonatal (0–24 h) mouse brains according to methods described previously (Hu *et al.*, 2013). Neurons were cultured in DMEM media (high-glucose) with fetal bovine serum (FBS, 10%), penicillin (100 U/mL) and streptomycin (100 μ g/mL) at 37 °C in a humidified 5% CO₂ atmosphere. Neurons were identified by immunostaining with the neuronal marker NF-M. After 14-d culture, neurons were fixed for 30 min in 4% paraformaldehyde, and then washed and rinsed with PBS. After that, they were blocked in PBS buffer containing 0.1% (w/v) Triton X-100 at room temperature (RT) for 30 min. Next, the neurons were incubated with anti-NF-M antibody (1:100) at 4 °C overnight. After rinsing with PBS, the neurons were overlaid with Cy3-conjugated secondary antibodies (1:200) for 1 h at RT. Neurons in the 24-hole plate were supplemented with 1% DAPI nuclear dye, covered with anti-fade mounting media, and viewed using a Nikon Eclipse E600 inverted fluorescence microscope (Japan). The NIH Image J 1.38 software (USA) was used to count cells. The numbers of neurons were quantified as the percentage of NF-M-positive cells to DAPI-labelled nuclei in 3 randomly selected fields of each well. Three wells for each group were counted.

Cellular model of AD and treatments. To study the neuroprotective effect of GP, neurons were divided into control group, GP group, model group (A β _{25–35} or H₂O₂) and treatment group (GP+ A β _{25–35} or GP+ H₂O₂). For treatment groups, GP (1, 10, 25, 50 and 100 μ g/mL) was added to neurons for 48 h after the

treatment with pre-aggregated 30- μ M A β _{25–35} (Hu *et al.*, 2013; Yu *et al.*, 2012) or 200- μ M H₂O₂ at 37 °C in a humidified 5% CO₂ atmosphere, and in all experiments except for the control group and GP group, A β _{25–35} and H₂O₂ were applied 24 h prior to treatment with GP. The A β _{25–35} was pre-aggregated as follows: 1 mg A β _{25–35} was dissolved in 940 μ L distilled water and incubated at 37 °C for 7 d in an incubator to induce aggregation. The aggregated A β _{25–35} was diluted to 500 μ M and stored at –70 °C (Hu *et al.*, 2013).

Cell viability assay. Neuronal viability was measured by the MTT assay using methods described previously (Lim *et al.*, 2015). The neurons in each group were plated in 96-well plates and treated with the MTT (5 mg/mL) for 4 h at 37 °C in the dark. Formed formazan crystals were dissolved in 150- μ L DMSO, and the optical densities (OD) were measured by an MR-96A microplate reader at 490 nm. The neuronal viability was expressed as a percentage of the non-treated control (Liu *et al.*, 2014).

Hoechst 33342 and propidium iodide (PI) double staining. Neuronal apoptosis was analysed by nuclear staining with Hoechst 33342/PI double staining as described in the literature (Wu *et al.*, 2013; Zhang *et al.*, 2013; Liu *et al.*, 2014). The neurons were fixed with 4% paraformaldehyde at RT for 1 h in 24-wells plates, and washed three times with PBS. Then, the neurons were incubated with Hoechst 33342 (5 μ g/mL, dissolved in PBS) at 37 °C for 10 min in the dark, and then incubated with PI (15 μ g/mL, dissolved in PBS) at 4 °C for 15 min. Next, the neurons were washed three times with PBS and observed by inverted fluorescence microscope (Japan). Apoptotic neurons were identified as distinctive condensed or fragmented nuclear structure (Fang *et al.*, 2012), and the PI positive neurons were counted by a cell counter. The apoptotic nuclei were counted in at least 200 neurons from three randomly selected fields in each group. Data are presented as the ratio of chromosomal condensation and morphological change as a percentage of total cells (Saeki *et al.*, 2011).

qRT-PCR analysis. Real-time quantitative polymerase chain reaction (qRT-PCR) was performed as previously described (Li *et al.*, 2016). The qRT-PCR was performed on an ABI 7700 fluorescence quantitative PCR system (USA) by using Power SYBR Green qPCR Master Mix. The PCR conditions were as follows: denaturation at 94 °C for 30 s, annealing at 60 °C

for 30 s and extension at 74 °C for 30 s, for a total of 40 cycles. All reactions were performed in triplicate. GAPDH was used as the control. The results were expressed as Ct values, and the relative changes in gene expression were determined by using a dual-standard curve method (Percopo *et al.*, 2014). The qRT-PCR primers used in this work as described in Table 1.

Rat model of AD and treatments. Female Wistar rats (260–280 g) were provided by the Liao Ning Chang Sheng Biotechnology Co., LTD, China. The experimental animal quality certification permit number was SCXK (Liaoning) 2010-0001. The Wistar rats were divided into three groups of eight each: the control group, model group (D-gal + Al, dissolved in normal saline) and GP group (D-gal + Al + GP 0.1 g/kg). The rats were intraperitoneally injected (i.p.) with 60 mg/kg D-gal and intragastrically (i.g.) administered with 40 mg/kg Al, once a day for 90 days. From day 60, the GP group was i.g. administered with 0.1 g/kg GP dissolved in deionized water, twice a day for 30 days. The control group was given an equal volume of vehicle daily for 90 days.

Animals were treated in compliance with the ‘guide for the care and use of laboratory animals’ published by the Shanghai Scientific and Technical Publishers of China in 2012. The experiments were performed with the approval of the Animal Ethics committee of Liaoning University of Traditional Chinese Medicine.

Morris water maze test. The apparatus (Chengdu TME Technology Co., Ltd., Chengdu, China) consisted of a circular pool (120 cm in diameter and 60 cm deep) with a black inner wall, which was conceptually subdivided into four equal quadrants and filled with water at 25 °C. After 30 days of treatment with GP, the Morris water maze was used to evaluate the spatial acquisition and memory ability of animals. The experiment includes a place navigation test and a spatial probe test. In the place navigation test, rats were released into different quadrants (North, South, East, West) to locate the escape platform for a maximum of 120 s. If rats were unable to reach the escape platform, they were guided to the platform where they had to stay for 30 s. The training trials were once a day for 5 consecutive days and were followed by the formal experiments on day 6. After a 24-h delay, the spatial probe test was performed. The platform was removed and the rats were allowed to freely swim for 120 s. Spatial learning ability was expressed as the time spent finding a platform. The swimming time, swimming path and time spent crossing the platform were recorded by a camera and automated analysing system (Kong *et al.*, 2015).

Table 1. The primers for qRT-PCR

Genes	Sense	Anti-sense	Size (bp)
Bcl-2	GTCCCGCCTCTTCACCTT	CCCACTCGTAGCCCTCT	409
Bax	GGCGAATTGGAGATGAAC	CCGAAGTAGGAGAGGAGG	307
Caspase-3	GAGCAGAGCCATGGGCACGT	AGGGCCCTGGCACACGGGAC	283
GAPDH	GACCACAGTCCATGCCATCAC	GCTGTTGAAGTCGCAGGAGAC	340

Elisa analysis. After the Morris water maze test, the rats were i.p. anaesthetized with 10% chloral hydrate, and the whole brain tissue was removed and the hippocampi were isolated from the brains, then a part of hippocampal tissues of each rat, respectively, was made into a 10% cold saline homogenate, and the remaining hippocampal and brain tissues, respectively, were stored in a freezer maintained at -70°C . The level of $\text{A}\beta_{1-42}$, p-tau and cAMP in the supernatant of the homogenate, respectively, were measured by the microplate reader according to the kit instructions, and the protein concentration was determined by BCA protein assay kit according to kit instructions.

Quantification of SOD, MDA, NOS and NO by kit. As mentioned above, the whole brain tissues (except hippocampus) of each rat, respectively, were made into 10% homogenates, and the levels of SOD, MDA, T-NOS, iNOS and nitric NO were measured by using commercial analysis kits.

Western blotting analysis. The western blotting analysis was performed as previously described (Li *et al.*, 2016). Briefly, a part of the hippocampus of each rat was lysed in RIPA buffer, the protein concentration was determined by BCA protein assay kit, and equal amounts of protein was separated by 10% SDS-PAGE and transferred onto polyvinylidene fluoride (PVDF) membranes. The membranes were blocked with 5% BSA in TBST buffer and were probed with primary antibodies anti-tubulin (1:1000), anti-PKA_C (1:1000), anti-p-PKA_C (1:1000), anti-CREB (1:500) and anti-p-CREB (1:500). Secondary horseradish peroxidase conjugated (HRP, 1:5000) anti-rabbit Ig G antibody in 5% nonfat milk blocking solution was then applied, and the immunoreactivity was visualized with ECL reagent. Densitometry was used to analyse the intensity of the resultant bands, and the relative absorbance of the bands was analysed by using Quantity One software (Bio-Rad).

Statistical analysis. The SPSS statistics software (IBM, Chicago, IL) was used to perform the data analysis. The one-way analysis of variance (ANOVA) with Kruskal–Wallis tests were used to evaluate differences among multiple groups. An unpaired, two-tailed Student's *t*-test was used to determine the statistical significance of two groups. All data were expressed as means $\pm$ SD of at least three independent experiments, and the level of significance was set at $p < 0.05$.

RESULTS

Molecular weight and purity of GP

The protein content of GP was determined by the coomassie brilliant blue kit, and the result was 78.2%. The subunit molecular weights of GP were determined

by SDS-PAGE, and the results showed that GP was composed of protein subunits of the molecular weights of 8, 15, 16, 18, 25, 27, 29, 65 and 66 kDa, of which the concentrations of 29 and 27 kDa protein subunits were the highest (Fig. 1 A). The molecular weight of natural GP was analysed by HPLC, and the results showed that the retention time of standard protein was 5.968 (97.4 kDa), 6.482 (66.2 kDa), 6.946 (43.0 kDa), 7.395 (31.0 kDa), 8.353 (20.1 kDa) and 9.379 min (14.4 kDa), respectively, and the retention time of the main absorption peak of GP was 6.546 min. As shown in Fig. 1 B, GP had two weak absorption peaks at the retention times of 5.867 and 9.412 min, the protein molecular weights were 80 and 0.6 kDa according to the linear relationship between the molecular weight of the standard protein and the retention time, and there was a strong absorption peak at the retention time of 6.546 min, which occupied more than 80% of the total peak area and the corresponding molecular weight was 65 kDa. It was suggested that there was a mixture composed of multiple protein subunits of molecular weights of approximately 65 and 80 kDa. The actual molecular weight of GP would be further confirmed in the near future.

Culture and assessment of neurons

Mouse cortical neurons were cultured and assessed with immunostaining of the neural marker NF-M. The quantitative analysis of immunostaining suggested that the purity of neurons was more than 95% (Fig. 2).

Effect of GP on neuronal viability by MTT assay

As shown in Fig. 3A, the viability of cells treated with GP within 1–100 $\mu\text{g}/\text{mL}$ showed no significant changes, while in the presence of $\text{A}\beta_{25-35}$ or H_2O_2 , GP rescued the neural viability in a dose-dependent manner (Fig. 3 B and C). The results also indicated that the survival rate of neurons in $\text{A}\beta_{25-35}$ or H_2O_2 model groups decreased significantly compared with the control group (57.11 $\text{A}\beta_{25-35}$, 61.24 H_2O_2 in model vs. control 100%,

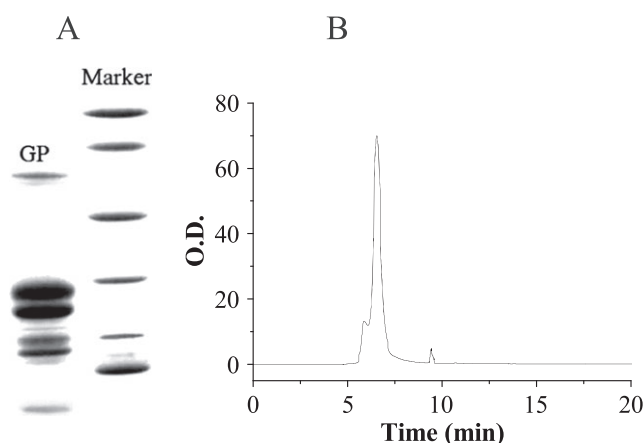


Figure 1. Molecular weight of GP. (A) The SDS-PAGE of GP. GP: ginseng protein. Marker: Protein Molecular Weight Marker, in descending order of their molecular weight was 97.4, 66.2, 43.0, 31.0, 20.1 and 14.4 kDa, respectively; (B) The HPLC of GP.

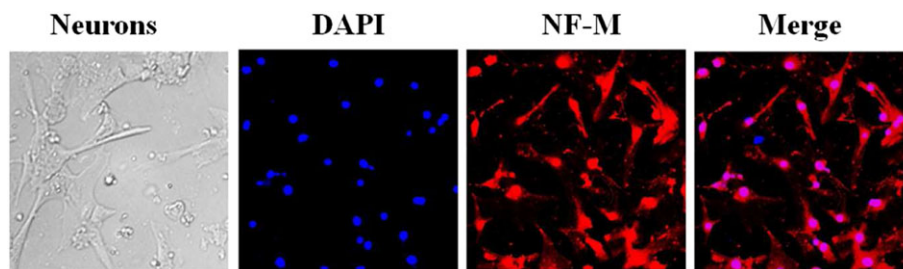


Figure 2. Characterization of neurons by immunostaining. The neurons were immunostained with anti-NF-M antibody (red); their nuclei were counterstained with DAPI (blue). Magnification: 400 times in phase contrast images, 200 times in immunofluorescence images. [Colour figure can be viewed at wileyonlinelibrary.com]

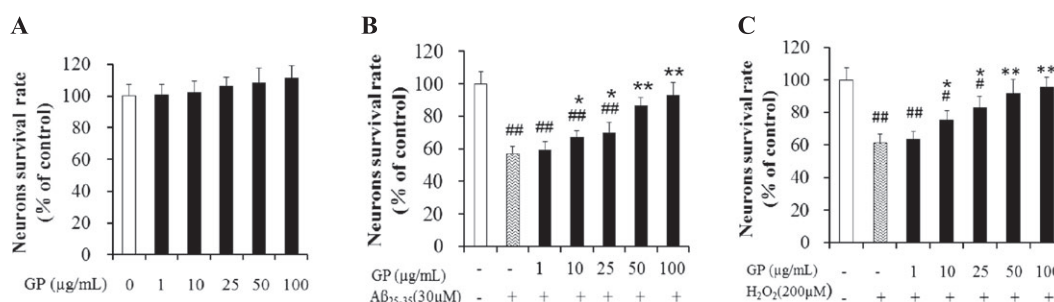


Figure 3. Effects of GP on neurons viability by MTT assay. (A) Effects of different concentrations of GP on non-treated neurons; (B) Effects of different concentrations of GP on Aβ₂₅₋₃₅-induced neurons; (C) Effects of different concentrations of GP on H₂O₂-induced neurons. Results were expressed as mean ± SD (*n* = 3). **P* < 0.05, ***P* < 0.01, vs. control; **P* < 0.05, ***P* < 0.01 vs. model (Aβ₂₅₋₃₅ or H₂O₂ groups).

P < 0.01), GP increased significantly compared with the model group (67.01, 70.03, 86.60, 92.78 or 75.26, 82.89, 91.75, 95.53 in GP 10, 25, 50, 100 µg/mL group vs. 57.11 or 61.24 in Aβ₂₅₋₃₅ or H₂O₂ model group, *P* < 0.05 or *P* < 0.01), and there was no difference between the high dose of GP (100 or 50 µg/mL group) and the control group (*p* > 0.05). Thus, 50 µg/mL of GP was chosen for the subsequently study.

Effect of GP on neuronal apoptosis by Hoechst 33342 and PI double staining

Neuronal apoptosis was detected by Hoechst 33342 and PI double staining. Treatment with GP alone did not cause neuronal apoptosis (Fig. 4 A, data analysis not shown), while in the presence of Aβ₂₅₋₃₅ or H₂O₂, GP significantly attenuated the loss of neurons and reversed the morphological changes (Fig. 4 B). The PI-positive cells (approximately 50%) in Aβ₂₅₋₃₅ (30 µM) or H₂O₂ (200 µM) groups significantly increased compared with the control group, while the number of PI-positive cells were significantly reduced when the neurons were treated with GP 50 µg/mL (Fig. 4 B). Data analysis, as shown in Fig. 4 C, indicates that the neuronal survival rate in Aβ₂₅₋₃₅ or H₂O₂ model groups were 47.83% and 54.33%, respectively (*p* < 0.001, vs. control). GP treatment improved the neuronal survival rate to 67.83% and 72%, respectively (*p* < 0.01, vs. model and vs. control). The counts of apoptotic bodies stained by Hoechst 33342 and PI suggested that GP significantly reduced nuclear condensation induced by Aβ₂₅₋₃₅ or H₂O₂, but there was significant difference compared with the control group.

Effect of GP on neuronal apoptosis by qRT-PCR

To investigate the molecular mechanism of GP on neuron survival, qRT-PCR was used to analyse the mRNA relative expression of Bcl-2, Bax and caspase-3. The relative expression of Bcl-2 mRNA in neurons treated by Aβ or H₂O₂ were 54.56% ± 4.91% and 62.98% ± 5.58%, respectively (Fig. 5 A, *p* < 0.01, vs. control). GP treatment increased the expression of Bcl-2 mRNA to 88.31% ± 7.17% and 92.02% ± 8.21% (Fig. 5 A, *p* < 0.01, vs. Aβ₂₅₋₃₅ or H₂O₂ model groups), respectively. The relative expression of Bax mRNA in Aβ₂₅₋₃₅ or H₂O₂ treated neurons were 217.25% ± 19.51% and 193.26% ± 19.36%, respectively (Fig. 5 B, *p* < 0.01 or *p* < 0.001, vs. control). GP treatment reduced the expression of Bax mRNA to 144.90% ± 14.20% and 122.58% ± 11.11% (Fig. 5 B, *p* < 0.01, vs. Aβ₂₅₋₃₅ or H₂O₂ model groups), respectively. And the ratio of Bax/ Bcl-2 mRNA in Aβ₂₅₋₃₅ or H₂O₂ model groups were 4.00 ± 0.36 and 3.13 ± 0.28, respectively (Fig. 5 C, *p* < 0.001, vs. control). GP treatment reduced the ratio of Bax/Bcl-2 mRNA to 1.64 ± 0.14 and 1.33 ± 0.12 (Fig. 5 C, *p* < 0.001, vs. Aβ₂₅₋₃₅ or H₂O₂ model groups), respectively. And the relative expression of caspase-3 mRNA in Aβ₂₅₋₃₅ or H₂O₂ model groups were 149.01% ± 14.61% and 134.96% ± 14.29%, respectively (Fig. 5 D, *p* < 0.01 or *p* < 0.001, vs. control). GP treatment reduced the expression of caspase-3 mRNA to 123.06% ± 12.50% and 113.15% ± 12.32% (Fig. 5 D, *p* < 0.05, vs. Aβ₂₅₋₃₅ or H₂O₂ model groups), respectively. Although there was still a significant difference between the GP treatment group and the control group (Fig. 5 B, C and D, *P* < 0.05, *P* < 0.01), GP was able to

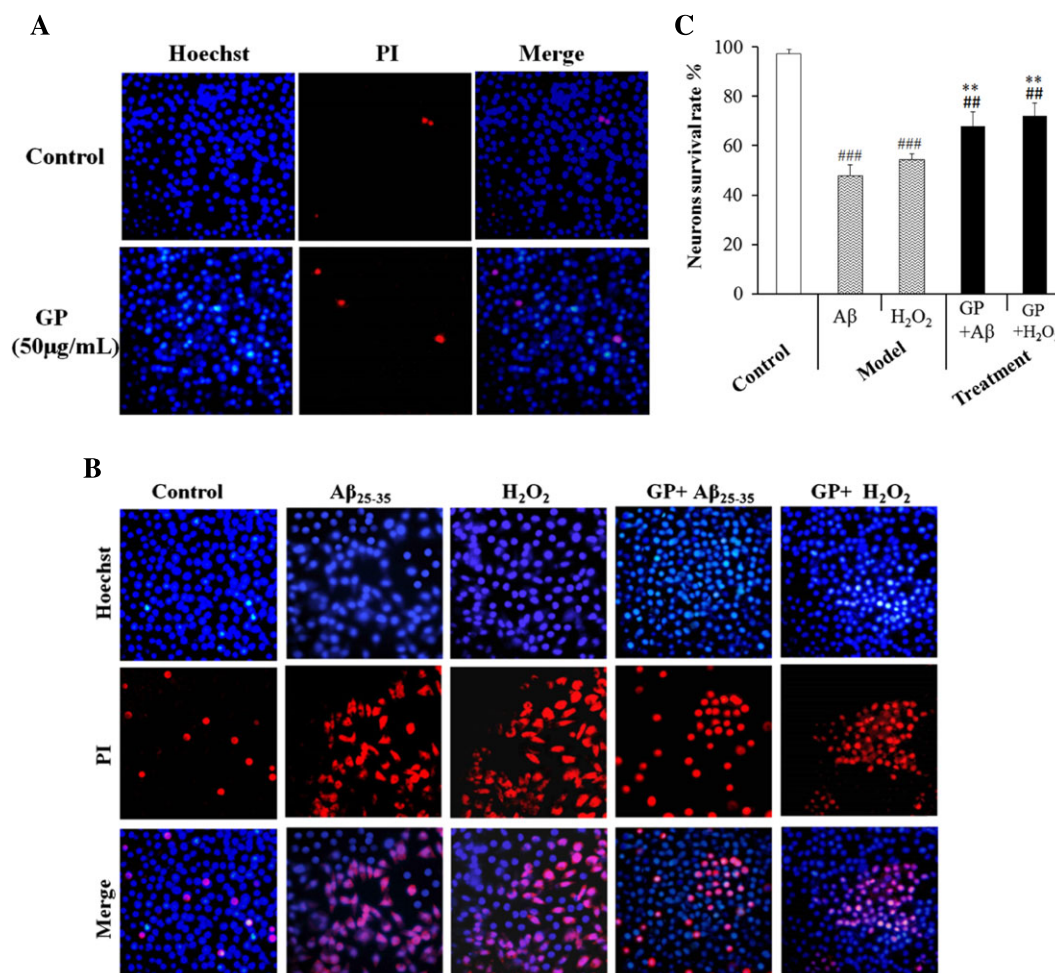


Figure 4. Effect of GP on neurons apoptosis by Hoechst 33342 and PI double staining. The photomicrographs of Hoechst 33342/PI double staining were shown in A and B; neurons survival percentage were analysed in C. (A) The representative images of control and GP group. (B) The representative images of control, model and treatment groups. PI positive cells (red, apoptosis cells); Hoechst 33342 counterstaining (blue, normal cells). (C) The neurons survival percentage was analysed by the Image J software. Magnification: 200 times. Results were expressed as mean $\pm$ SD ($n = 3$). $^{##}P < 0.01$, $^{###}P < 0.001$, vs. control; $^{*}P < 0.01$ vs. model. [Colour figure can be viewed at wileyonlinelibrary.com]

restore the relative expression of Bcl-2, Bax and caspase-3 mRNA in the GP+H₂O₂ groups to normal levels (Fig. 5 A, B and D, $p > 0.05$, vs. control).

Effect of GP on the ethology of AD rats

The learning and memory of rats were evaluated by a Morris water maze test. The experimental results of place navigation (Fig. 6 A, B and C) showed that the model group significantly prolonged the escape latency compared to the control group (40.24s in model group vs. 16.28s in control group, $P < 0.01$, Fig. 6 C), and the GP group significantly shortened the escape latency compared to the model group (18.78s in GP group vs. 40.24s in model group, $P < 0.01$, Fig. 6 C). The experimental results of the spatial probe (Fig. 6 D and E) showed that the model group significantly decreased the platform crossing times compared to the control group (4.88 times in model group vs. 6.25 times in control group, $P < 0.01$), the GP group significantly increased that compared to the model group (5.75 times for the GP group vs. 4.88 times for the control group, $P < 0.05$, Fig. 6 D). Also, the percentage of time spent at the

original platform quadrant was significantly different among the three groups. The model group significantly decreased the percentage of time compared to the control group (23.11% in the model group vs. 30.25% in the control group, $P < 0.01$), and the GP group significantly increased the percentage of time compared to the model group (26.33% in GP group vs. 23.11% in model group, $P < 0.05$, Fig. 6 E). The results suggested that GP could enhance the cognitive performance of rats but could not completely recover cognition to the normal level.

Effect of GP on the levels of A β_{1-42} and p-tau in AD rats

The abnormal deposition of A β and tau-containing neurofibrillary tangles are the major pathological hallmarks of AD (Koo *et al.*, 2013). Elisa analysis was used to determine the levels of A β_{1-42} and p-tau in order to evaluate the effect of GP on pathological damage in AD rats. The results showed that the levels of A β_{1-42} and p-tau in the hippocampus of the model group and the control group were significantly different (2.95 A β_{1-42} , 3.71 p-tau in model group vs. 2.57 A β ,

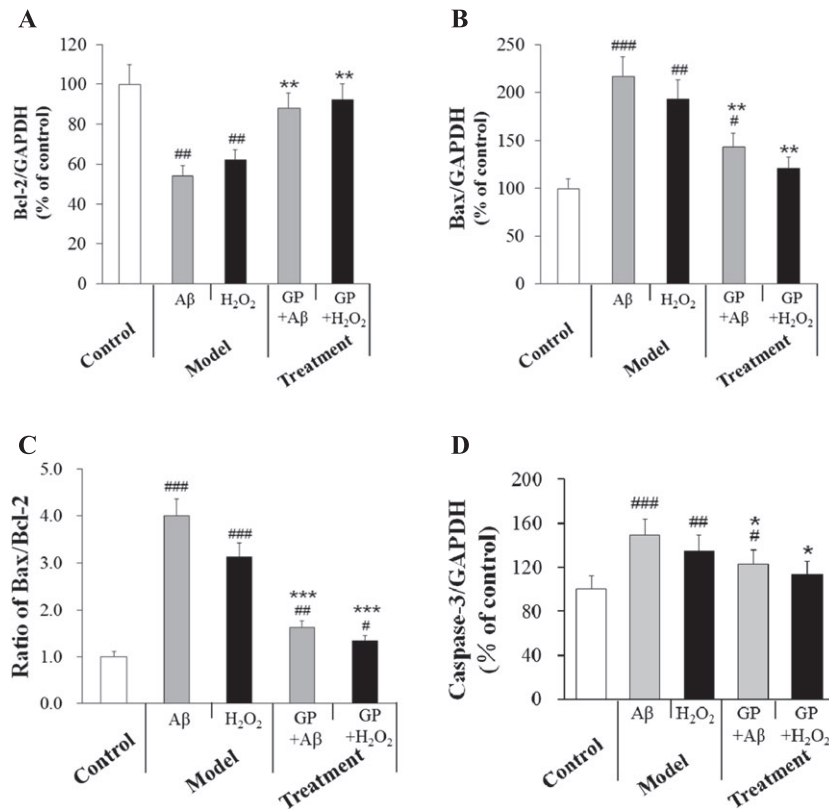


Figure 5. Effect of GP on neurons apoptosis by qRT-PCR. (A) The relative expression of Bcl-2 mRNA; (B) the relative expression of Bax mRNA; (C) the ratio of relative expression of Bax/ Bcl-2 mRNA; (D) the relative expression of caspase-3 mRNA. [#] $P < 0.05$, ^{##} $P < 0.01$, ^{###} $P < 0.001$, vs. control; ^{*} $P < 0.05$, ^{**} $P < 0.01$, ^{***} $P < 0.001$, vs. model.

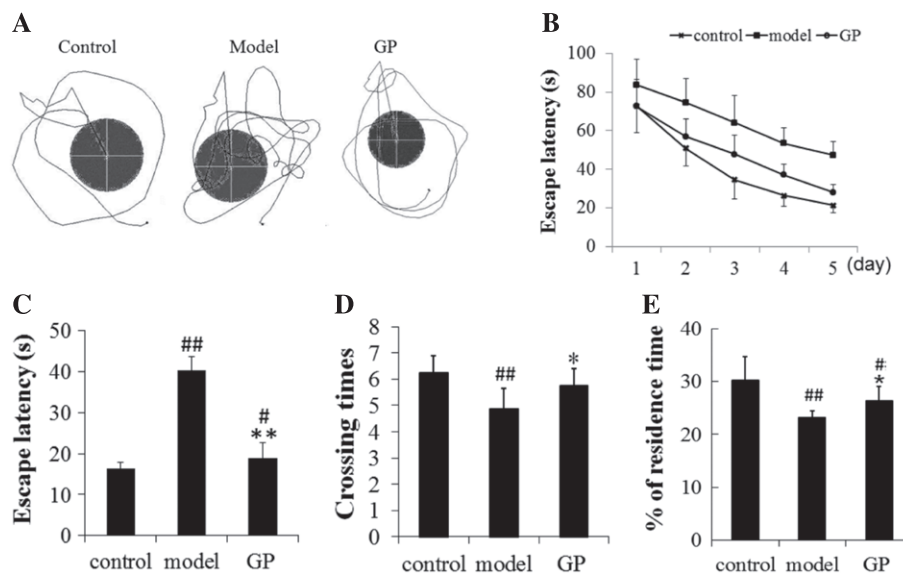


Figure 6. Morris water maze test. The experimental results of place navigation are shown in A, B and C, and the results of spatial probe are shown in D and E. (A) Representative swimming paths of each groups (at day 6); (B) escape latency in the training period (1–5 days); (C) escape latency in the formal experiments (on 6th day); (D) the number of times of crossing the target quadrant; (E) the percentage of residence time in the target quadrant. The values were expressed as the mean $\pm$ SD. [#] $P < 0.05$, ^{##} $P < 0.01$, vs. control; ^{*} $P < 0.05$, ^{**} $P < 0.01$, vs. model.

3.33 p-tau in control group, $P < 0.05$ or $P < 0.01$, Fig. 7 A and B), and the GP group significantly decreased the levels of A β_{1-42} and p-tau compared to that of the model group (2.59 A β_{1-42} , 3.42 p-tau in GP group vs. 2.95 A β_{1-42} , 3.71 p-tau in model group, $P < 0.05$ or

$P < 0.01$, Fig. 7 A and B). The results in Figs 6 and 7 showed that using D- gal combined with Al treated rats for 90 days induced an AD-like rat model, while GP could reduce the behavioural and pathological changes of the AD rats.

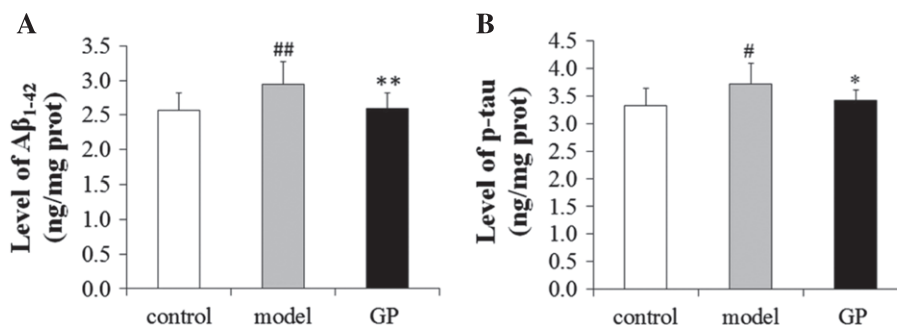


Figure 7. Levels of Aβ₁₋₄₂ and p-tau. (A) Aβ₁₋₄₂ levels; (B) The p-tau levels. Values are expressed as mean ± SD. [#]*P* < 0.05, ^{##}*P* < 0.01, vs. control; ^{*}*P* < 0.05, ^{**}*P* < 0.01, vs. model.

Effect of GP on the oxidation indexes in the brain tissues of AD rats

To evaluate the antioxidant capacity of GP on AD rats, the SOD, MDA, T-NOS, iNOS and NO levels of the whole brain tissue without hippocampus of AD rats were analysed by the kit. As shown in Fig. 8 A–F, GP significantly improved the activity of SOD (212.50 in GP group vs. 173.88 in model group, *p* < 0.01), reduced the level of MDA, NO, T-NOS and iNOS (6.56 MDA, 3.80 NO, 8.13 T-NOS, 6.01 iNOS in GP group vs. 7.44 MDA, 4.44 NO, 10.91 T-NOS, 7.78 iNOS in model group, *p* < 0.05 or *p* < 0.01), and the activity of NOS was not significantly different compared with the control (8.13 T-NOS and 6.01 iNOS in GP group vs. 7.90 T-NOS and 5.81 iNOS in control group, *p* > 0.05).

Effect of GP on protein expression in the hippocampus of AD rats

To investigate the molecular mechanism of GP on cognitive improvement in rats, an Elisa test was used to

analyse the cAMP concentration (Fig. 9 A), and western blotting was used to analyse PKA, p-PKA, CREB and p-CREB protein expression (Fig. 9 B–H). The results showed that GP significantly enhanced the concentration of cAMP (0.23 in GP group vs. 0.18 in model group, *p* < 0.05, Fig. 9 A), and the concentration was not significantly different than the control group (0.23 in GP group vs. 0.31 in control, *p* > 0.05, Fig. 9 A). And GP enhanced the protein expression of PKA (89.39 in GP group vs. 69.02 in model group, *p* < 0.05, Fig. 9 C), p-PKA (144.19 in GP group vs. 91.01 in model group, *p* < 0.01, Fig. 9 D), the ratio of p-PKA/PKA (1.61 in GP group vs. 1.32 in model group, *p* < 0.05, Fig. 9 E), and the protein expression of p-CREB (74.64 in GP group vs. 22.65 in model group, *p* < 0.01, Fig. 9 G), the ratio of p-CREB/CREB (0.71 in GP group vs. 0.24 in model group, *p* < 0.01, Fig. 9 H), but the ratio of p-CREB/CREB did not recover to the normal level (0.71 in GP group vs. control, Fig. 9 H).

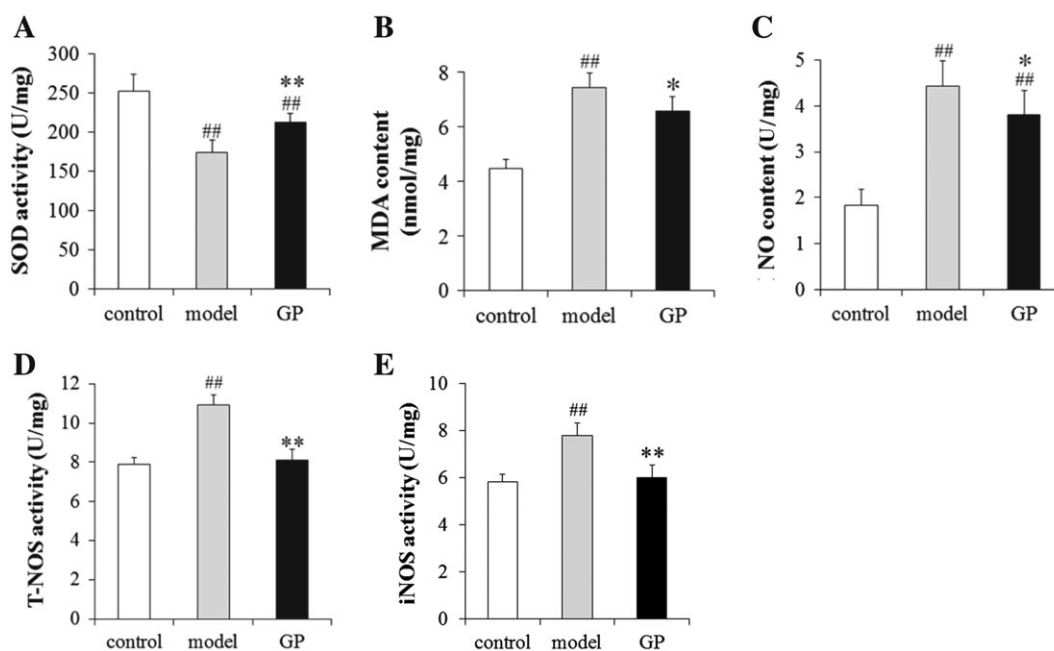


Figure 8. The levels of SOD, MDA, NOS and NO in the brain tissues. (A) Activity of SOD; (B) content of MDA; (C) content of NO; (D) activity of T-NOS; (E) activity of iNOS. Values are expressed as mean ± SD. ^{##}*P* < 0.01, vs. control; ^{*}*P* < 0.05, ^{**}*P* < 0.01, vs. model.

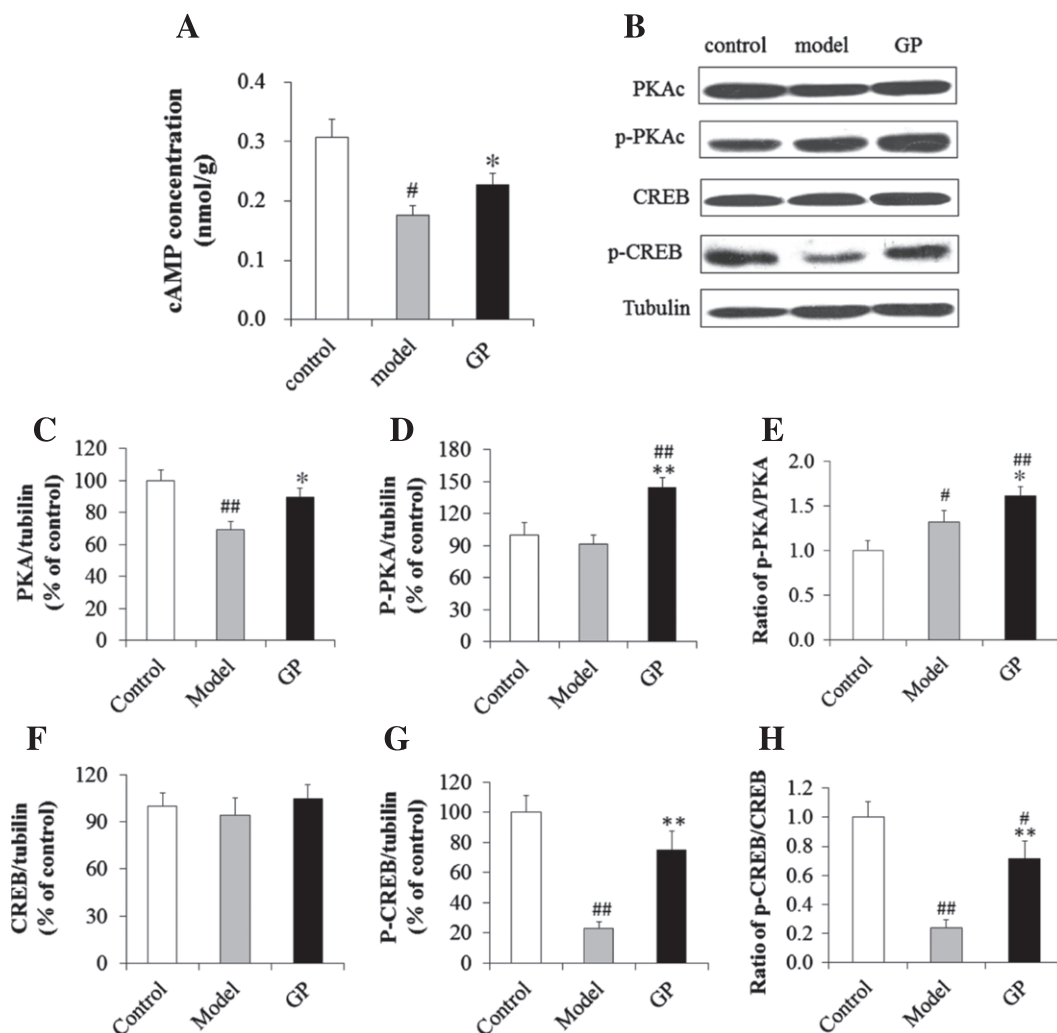


Figure 9. The cAMP, PKA and CREB levels in the hippocampus of AD rats. (A) Concentration of cAMP; (B) representative image of target protein levels; (C) relative expression of PKA protein; (D) relative expression of p-PKA protein; (E) relative expression ratio of p-PKA/PKA protein; (F) relative expression of CREB protein; (G) relative expression of p-CREB protein; (H) relative expression ratio of p-CREB/CREB protein. Values are expressed as mean $\pm$ SD. [#] $P < 0.05$, ^{##} $P < 0.01$, vs. control; ^{*} $P < 0.05$, ^{**} $P < 0.01$, vs. model.

DISCUSSION

With increasing life expectancy, AD is increasing more and more rapidly, and it has become a major health issue and an important social problem in the world (Brookmeyer *et al.*, 2007). Modern pharmacological studies have indicated that oxidative stress is an important pathological event in the incidence of early AD (Aliev *et al.*, 2014).

In this study, the cellular and rat models of AD were established by A β_{25-35} , H $_2$ O $_2$ and the combination of D-gal with Al. A β and H $_2$ O $_2$ induced neurotoxicity has been widely used for researching the effect of neuronal responses to oxidative stress. Excessive D-gal triggers reactive oxygen species (ROS) formation in the brain and later induces oxidative stress (Ali *et al.*, 2015; Lu *et al.*, 2007; Lu *et al.*, 2009). Injection of D-gal can induce brain ageing and accelerates senescence to AD-like symptoms (Ali *et al.*, 2015). Al is a ubiquitous environmental poison. Human beings are vulnerable to Al toxicity (Flaten 2001; Gonda and Lehotzky, 1996). Al has neurotoxicity (Kumar and Gill, 2009; Poirier *et al.*, 2011; Wu *et al.*, 2012), and it can accumulate in different

brain areas (Yokel *et al.*, 2001; Zatta *et al.*, 2002), including the cortex, hippocampus, corpus callosum, etc. (Platt *et al.*, 2001; Flaten, 2001). Epidemiological studies have indicated that Al may lead to cognitive dysfunction and learning and memory impairment (Zhang *et al.*, 2014). A wealth of evidence suggests that excessive intake of aluminium is a major risk factor for AD (Zhang *et al.*, 2014).

To investigate the antioxidant capacity of GP on AD rats, the SOD, MDA, T-NOS, iNOS and NO level of the brain tissues of AD rats were analysed by the kit. The activities of SOD directly reflect the ability of scavenging oxygen free radicals (Shao *et al.*, 2016). MDA is a by-product of lipid peroxidation induced by free radicals and is also widely used as a biomarker of oxidative stress (Pan *et al.*, 2013). NOS, in particular inducible NOS (iNOS) and the production of NO are observed in AD tissue as increased amounts of nitrotyrosine-modified proteins (Hensley *et al.*, 1996).

Loss of memory and cognitive dysfunction are the major pathological characteristics of AD (Koo *et al.*, 2013). The cAMP/PKA/CREB pathway is involved in the formation of memory (Suzuki *et al.*, 2011). Protein kinase A (PKA) activity relies on PKA catalytic subunit

(PKAc) phosphorylation, and ROS is a determined factor for disturbing the phosphorylation of PKAc (Du *et al.*, 2014; Humphries *et al.*, 2005). The PKA activated by cAMP can be phosphorylated by the cAMP response element binding protein (CREB) (Gonzalez and Montminy, 1989). Phosphorylated CREB interacts with the transcriptional coactivator CREB-binding protein (CBP) to regulate the transcription of an array of pro-survival and anti-apoptotic genes, which are required for the synaptic plasticity that mediates long-term potentiation (Lonze and Ginty, 2002), and the CREB pathway is involved in the formation of short-term memory (Suzuki *et al.*, 2011).

Ginseng is the root and rhizome of *P. ginseng* C. A. Mey, as a traditional herbal medicine, it has been used in China for more than 2000 years. Ginseng has many pharmacological effects, such as anti-oxidation, anti-ageing and enhanced immunity (Wesnes *et al.*, 2000; Hsieh *et al.*, 2000; Gold *et al.*, 2001). Ginseng's effects on improving cognition have been recorded in 'Ben Cao Gang Mu', which is by far the most comprehensive herbal book compiled during the Ming Dynasty in China by the great medical scientist Shi-Zhen Li. The chemical components of ginseng include ginsenosides, polysaccharides, proteins, peptides, and so forth (Ru *et al.*, 2015). At present, ginsenosides are considered as the main bioactive constituents of ginseng, and there are few studies on the proteins and polypeptides. O'Keefe reported that the protein components extracted from natural products represented a largely untapped source of potential bioactive molecules (O'Keefe, 2001). It was reported that GP had the effect of an antioxidant, enhanced immunity, anti-radiation, anti-stress, etc. in the articles from the following Chinese core journals: *Food Science and Technology* (2012), 3, 79–82; *Traditional Chinese Drug Research and Clinical Pharmacology* (2011), 2, 138–141; *Lishizhen Medicine and Materia Medica Research* (2010) 9, 2143–2144; *Science and Technology of Food Industry* (2011), 11, 406–407; and so forth. Our previous studies also showed that GP had a neuroprotective effect (Li *et al.*, 2016; the Chinese core journals: *Journal of Jilin University Medicine Edition*, 2013, 6:1138–1142 and *Chinese Journal of Hospital Pharmacy*, 2015, 35: 1906–1910), which could improve cognitive function and reduce the pathological damage in AD rats induced by D-gal and Al (Li *et al.*, 2016), but its mechanism was not very clear. Consistent with our previously published article (Li *et al.*, 2016), this study still utilizes oral administration. Although the absorption mechanism of GP *in vivo* has not been clear, and it was speculated that GP was hydrolysed into low molecular peptides and a small amount of amino acids, which was described in our previous articles published in the Chinese core journals (*Modern*

Chinese Medicine, 2016, 18: 6–10); or GP regulated body functions such as enhancing immunity and antioxidation playing the role of anti-AD (Li *et al.*, 2016). For the oral absorption mechanism of GP, the further research work is ongoing.

In this paper, we studied the effect of GP on cells and animals with AD *in vitro* and *in vivo*. The experimental results showed that the GP could improve the survival rate of neurons and reduce neuronal apoptosis *in vitro*. GP could improve learning and memory, SOD activity, cAMP concentration, the ratio of protein expression of p-PKA/PKA and p-CREB/CREB, and reduce the pathological damage, the NOS activity, MDA and NO content of AD rats whose damage was induced by D-gal and Al *in vivo*. It has been suggested that GP has neuroprotective and anti-AD effects *in vitro* and *in vivo*, and the mechanism is in part mediated by antioxidant and apoptosis-inhibiting effects, which may be associated with the cAMP/PKA/CREB pathway.

It has been reported that any stress, neurotransmitters, growth factors, glutamate and membrane depolarization could activate the intracellular signaling pathway and cause the activation of the CREB-dependent cascade, which is triggered by post-translational modifications such as phosphorylation of CREB (Alberini 2009; Tronel *et al.*, 2005). CREB is considered to be a central activator that could integrate a variety of stimuli (Alberini 2009). Our previous study suggested that the neuroprotective effect of GP was in part mediated by PI3K/Akt pathway activation. As reported, the pathway of cAMP/PKA and the PI3K/Akt converge on CREB (Alberini 2009). Thus, it could be hypothesized that, the anti-AD mechanism of GP likely activates the CREB pathway through multiple channels, and then the downstream apoptotic family. Subsequently, a study on the detailed mechanism of the neuroprotective effect of GP should be conducted.

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Conflict of interest

On behalf of all authors, the corresponding author states that there is no conflict of interest.

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